Edouard Butaye

Research scientist / CFD Developer

+33 769175524
ebutaye@sfr.fr
edouard-butaye
Edouard-Butaye

Skills

CFD Development: `C/C++`, `git`

Automation: `Bash`

Post-processing: `Python`, LaTeX

CFD software: `TRUST/Trio_CFD`, `Code_Saturne`, `Neptune_CFD`, `SYRTHES`

Mesh generation: `Salomé (mesh)`, `MMG`

Visualization software: `ParaView`, `VisIt`, `PyVista`, `Jupyter`

Languages

French: Mother tongue

English: TOEIC (865)

Spanish: Notions

Chinese: HSK 2 (A2)

Extracurriculars

Volunteering: student humanitarian associations, Red-Cross, Food Banks

Sports: Volleyball (regional level), Fitness, Running

Research

2025 - Postdoctoral researcher — CEA - Saclay (France)

Hybrid RANS-LES modelling.

A. Burlot

Implementation of the hybrid RANS-LES method *Hybrid Temporal Large Eddy Simulation* (HTLES) developed by Rémi Manceau in `TRUST/Trio_CFD` (`C++`) software. A turbulent forcing term is introduced into the momentum equation in order to address the grey-area issue that arises during the transition from RANS to LES. Creation of validation forms and test cases to ensure non-regression. **High Performance Computing**. GPU porting with Kokkos.

Objective: Define a physical criterion for mesh refinement to transition to LES in areas of interest. This criterion must be independent of the mesh and depend solely on physical quantities from a RANS simulation.

Test cases: Backward facing step, periodic hill, T-junction.

2021 - 2024 PhD student in fluid mechanics — PROMES - CNRS - Perpignan (France)

Particle-Resolved Direct Numerical Simulation (PR-DNS) of anisothermal fluid-solid particles flows.

A. Toutant, S. Mer, F. Bataille

Development in `TRUST/Trio_CFD` (`C++`) software. Creation of validation forms and test cases to ensure non-regression.

PR-DNS of anisothermal fluidized bed ($8\,500+$ solid particles). One-fluid formulation associated with the **Front-Tracking** method.

High Performance Computing ($> 3\,800$ CPUs).

Post-processing: `GNU Parallel`, `VisIt` and `PyVista`. Computation of Lagrangian and Eulerian statistics with `Python` library `Pandas` (multi-processing).

Teaching: $92\,h$ of supervised TD/TP at Univ. Perpignan.

Supervision: 3 internships (2-5 months).

2021 - 2021 Master 2 internship (6 months) — PROMES - CNRS - Perpignan (France)

Multiscale gas-liquid simulation of horizontal boiling flows in concentrated solar power plants driven by direct steam generation using `Neptune_CFD`.

A. Toutant, S. Mer, F. Bataille

Study of horizontal **turbulent boiling two-phase** flows. Coupling of `Neptune_CFD` with `SYRTHES` to resolve conduction in the wall of the receiving tube. **High Performance Computing**.

Teaching

2022-2024 Teaching assistant (92h) — University of Perpignan (France)

Fluid Mechanics & heat transfer 24h Tutorial Classes (TC), 12h Laboratory Sessions (LS): heat transfer coefficient, Nusselt number, boundary layers, Bernoulli's principle, pressure losses

Numerical tools 18h TC: discretization of partial differential equations using the finite difference method

Wind Energy 24h TC: wind turbine production assessment with WindPro software, Blade element theory (computation of lift, drag, axial thrust on NACA airfoils)

Radiative transfer 13.5h TC: luminance, emittance, radiosity, form factors, grey and black bodies, electrical analogy

Studies

2018 - 2021 Engineering school — ISAE-ENSMA - Poitiers (France)

Fluid mechanics, turbulent flows, two-phase flows, aerodynamics, thermal modelling, numerical simulation

2020 - 2021 Master 2 of research — ISAE-ENSMA (PPRIME) - Poitiers (France)

Thermal specialisation: nano-transfer and inverse methods

2016 - 2018 Preparatory courses (Maths & Physics) — Kerichen - Brest (France)

Two-year undergraduate intensive preparation courses for the entrance exams to top French engineering schools

Publications

- **E. Butaye**, S. Mer, F. Bataille, A. Toutant – Extensive analysis of hydrodynamics in liquid-solid fluidized beds with Particle-Resolved Simulations, Int. J. Multiph. Flow, 2026.
- **E. Butaye**, R. Quintana, S. Mer, F. Bataille, A. Toutant – Investigation of heat transfers with particle-resolved simulations: from Stokes flow to fluidized bed – Int. J. Heat Mass Transf., 2025
- **E. Butaye**, A. Toutant, S. Mer, F. Bataille – "Development of Particle Resolved - Subgrid Corrected Simulations : Hydrodynamic force calculation and flow sub-resolution corrections" – *Comput. & Fluids.*, 2023.
- **E. Butaye**, A. Toutant, S. Mer – "Euler-Euler multi-scale simulations of internal boiling flow with conjugated heat transfer" – *Appl. Mech.*, 2023.

Conferences

- **E. Butaye**, A. Burlot, "Hybrid RANS-LES simulation: definition of a remeshing criterion based on physical considerations", DLES15, Delft, May 2026.
- G. Ologaray, **E. Butaye**, A. Toutant, S. Mer, F. Bataille. "Simulations d'écoulements anisothermes gaz-particules appliqués aux centrales solaires à concentration (CSP)". *32^{e} Congrès Français de Thermique* - Strasbourg - Jun. 2024.
- **E. Butaye**, R. Quintana, A. Toutant, S. Mer, F. Bataille. "Particle-Resolved Simulation of anisothermal fluidized beds". *9th Int. Conf. on Multiphase Flow and Heat Transfer (ICMFHT)* - London - Apr. 2024.
- **E. Butaye**, S. Mer, A. Toutant, F. Bataille. "Direct numerical simulation of fluid-solid particles flows using front-tracking approach". *IUTAM Symposium: From Stokesian suspension dynamics to particulate flows in turbulence*, Sept. 2022 - Toulouse, France.
- **E. Butaye**, M. Ploquin, S. Mer, A. Toutant, F. Bataille. "Euler-Euler multi-scale simulations of internal turbulent boiling flow with conjugated transfer". *Dispersed two-phase flow - SHF, Oct. 2021 - Online Event.*

Seminar

- **E. Butaye**, A. Burlot. "Modélisation hybride RANS/LES". CEA-ISAS Annual Scientific Conference - Saclay (France) - Jun. 2026.
- **E. Butaye**, A. Toutant, S. Mer, F. Bataille. "Simulations anisothermes d'écoulements fluide-particules solides". CEA-ISAS Annual Scientific Conference - Saclay (France) - Jun. 2025.
- **E. Butaye**, A. Burlot. "Hybrid RANS-LES modelling". *6th seminar `TrioCFD`* - Massy (France) - Oct. 2025.
- **E. Butaye**, A. Toutant, S. Mer, F. Bataille. "Particle-Resolved Simulations of fluid-solid particles flows with `TrioCFD`". *Departmental seminar* - Saclay (France) - Apr. 2025.
- **E. Butaye**, A. Toutant, S. Mer, F. Bataille. "Particle-Resolved Simulations (PRS) of anisothermal fluid-solid particles flows". *5th seminar `TrioCFD`* - Massy (France) - Oct. 2023.

Review

- International Journal of Heat and Mass Transfer : 2